

BRAC University
CSE340: Computer Architecture
Quiz#4

Duration : 20 Minutes

Total Marks : 20 Set: Abacus

ID:

Name:

Sec:

1. Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	0	111	Data
011	1	001	Data
100	0		
101	1	000	Data
110	1	010	Data
111	0		

Memory accesses:

1. 36
2. 58
3. 22
4. 36
5. 62

- a. For the above memory access, determine if it will be a hit or a miss. [5]
- b. Calculate the hit and miss ratio. [4]
- c. Why do we add layers to the memory hierarchy? What is the disadvantage of adding more layers? [4]

2. Let's say we have 32 bit address space. we have a cache where we can store 1 byte data per block in cache (1 byte/block) . We have 20 bit space for storing tag bits for each entry in cache. What is the total size of the cache (considering tag, valid bit and data block) ? [4]

Index	Valid bit	Tag (20bit)	Data (1 byte)
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3. What is the write through method? What problem arises in this method and how do we solve it? [3]

BRAC University
CSE340: Computer Architecture
Quiz#1

Duration : 20 Minutes

Total Marks : 20

Set: Byte

ID:

Name:

Sec:

1. Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	1	111	Data
011	1	001	Data
100	0		
101	1	000	Data
110	0	010	Data
111	0		

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2. 58
3. 22
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